PHP with MySQL — Student Notes

Class 1: Introduction, Evolution & PHP Versions

1. What Actually Is PHP?

PHP stands for "PHP: Hypertext Preprocessor" (yes, the acronym contains itself — that's called a recursive acronym, and it's a small programming-culture joke you'll now understand). In plain terms:

> 💡 *PHP is a server-side scripting language used to build the parts of a website that need to think, decide, or fetch data before showing you a page.*

Let's ground this with something every student already understands — ordering food at a restaurant.

Real-World Analogy: The Restaurant

- You (the customer) = the Browser (Chrome, Firefox, etc.)
- The Menu you look at = the Webpage (HTML/CSS you see)
- The Waiter who takes your order and brings food = the Request/Response cycle
- The Kitchen, where the real cooking happens out of your sight = the Server, running PHP

When you order "Chicken Biryani," you don't cook it yourself, and you don't see the cooking process — you just place an order and a plate arrives. Similarly, when you log into Instagram, you don't see PHP (or whatever backend language) checking your password against a database — you just place a "request" (click Login) and get a "response" (your feed).

This is exactly why PHP is called server-side: the code runs on the kitchen (server), not on your dining table (browser). Your browser never sees the PHP code itself — only the final HTML "dish" it produces.

2. Why Does PHP Exist? A Short History

PHP wasn't designed in a boardroom by a company planning to conquer the web. It was built by one person solving one personal, boring problem — and that's worth knowing, because it explains why PHP feels practical rather than academic.

The Origin Story

In 1994, a programmer named Rasmus Lerdorf wanted to know who was viewing his online résumé. So he wrote a small set of CGI (Common Gateway Interface) scripts in C to track visitors. He called this toolkit "Personal Home Page Tools" — PHP Tools. That's it. That's the origin of a language that now powers roughly 3 out of every 4 websites whose backend language is known.

> 💡 *Real-world parallel: Gmail started as an internal 20%-time project at Google. Post-it Notes were a failed "super glue" experiment. PHP started as one guy's visitor counter. Useful tools often start small and unglamorous.*

How It Evolved Into a Real Language

Other developers found Rasmus's tools useful and started requesting features — form handling, database connections, and so on. Slowly, "a set of scripts" turned into "a language." Here's the version timeline you should remember, especially for the interview questions later in this note:

Version	Year	Key Change	Why it mattered
PHP/FI 1.0 & 2.0	1994–1997	Rasmus Lerdorf's "Personal Home Page Tools"	A personal CGI script to track visits to his online résumé — not even meant to be a language
PHP 3	1998	Rewritten as a real language with an extensible core	First version people actually called "PHP"; parser rewritten by Zeev Suraski & Andi Gutmans

Version	Year	Key Change	Why it mattered
PHP 4	2000	New engine: the Zend Engine	Big performance and stability jump; powered the first PHP boom (early 2000s web)
PHP 5	2004	Proper Object-Oriented Programming (Zend Engine II)	Let PHP build large, structured applications — enabled frameworks like early Laravel/Symfony ancestors
PHP 7	2015	Massive performance rewrite (PHPNG engine), scalar type hints	Roughly 2x faster than PHP 5 with less memory — why WordPress/Facebook-scale sites stayed on PHP
PHP 8	2020 onward (8.0–8.4)	JIT compiler, union types, named arguments, match expression, enums, readonly properties	Modern PHP that competes with newer languages while keeping backward compatibility

3. Where Is PHP Actually Used Today?

A common student doubt: "Is PHP outdated? Should I learn Python/Node instead?" Let's answer this with facts, not opinions.

- WordPress (built in PHP) powers over 40% of all websites on the internet — every WordPress blog, news site, or small business site you've visited runs PHP under the hood.
- Facebook was originally built in PHP, and even today runs a heavily modified version of it called Hack/HHVM.
- Wikipedia's MediaWiki software is written in PHP.
- Large parts of platforms like Etsy and Slack's early infrastructure used PHP.

> 💡 *Interview-relevant fact: "Is PHP still relevant?" is a common opening interview question. Your answer: PHP 8's JIT compiler and performance rewrites in PHP 7 keep it fast, it still runs a huge share of the web, and frameworks like Laravel keep it modern and in active demand for backend/full-stack roles.*

4. Static vs Dynamic Websites (A Preview)

You'll study this properly in Class 2, but here's the seed idea, because it's the whole reason PHP exists:

- Static website: the HTML file sent to your browser is exactly the same for every visitor. A digital pamphlet — same content for everyone, every time.
- Dynamic website: the HTML sent to your browser is generated freshly, often differently, for each visitor. Your Amazon homepage looks different from your friend's, because it shows your order history, your recommendations, your cart.

PHP is the tool that generates that per-visitor HTML on the server before sending it out. That's the whole job of this language, and everything you learn this semester builds toward doing that well and safely.

5. A First Glance at PHP Code

We haven't covered syntax yet (that's Class 3), but seeing one tiny real example now will make the history above click into place. This is what a PHP file looks like when it's embedded inside HTML:

```
<!DOCTYPE html>
<html>
<body>

  <h1>Welcome!</h1>

  <?php
```

```
    // This part runs on the SERVER, before the page is sent to you
    $visitorName = "Rahul";
    echo "Hello, " . $visitorName . "! You are visitor number 42.";
  ?>

</body>
</html>
```

Notice the `<?php ... ?>` block sitting right inside normal HTML. Everything outside those tags is sent to the browser as-is. Everything inside those tags is executed on the server — the browser never sees the PHP code, only the word "Hello, Rahul! You are visitor number 42." that it produces. This is exactly the kitchen-vs-dining-table idea from Section 1.

6. In-Class Discussion (Practice Activity)

Name three websites you personally use every day, and try to guess: does the content change based on who's logged in? If yes, something on a server is almost certainly generating that page dynamically — possibly with PHP, possibly with another server-side language (Python/Django, JavaScript/Node, Java, Ruby). The concept is universal even when the language changes.

7. Homework

Read a short article on "why PHP is still relevant in 2026" and come prepared to share one fact you didn't know.

8. Interview Questions From This Class

Question	Answer
What does PHP stand for?	PHP: Hypertext Preprocessor — a recursive acronym (the P stands for PHP itself).
Who created PHP and when?	Rasmus Lerdorf, in 1994, originally as "Personal Home Page Tools" to track visits to his résumé.
Is PHP a client-side or server-side language? Why?	Server-side — the code executes on the web server, and only the resulting HTML output is sent to the browser. The client never sees the PHP source.
Which PHP version introduced proper OOP support?	PHP 5, via the Zend Engine II.
What was the major improvement in PHP 7?	A rewritten engine (PHPNG) that roughly doubled performance and reduced memory usage compared to PHP 5, plus scalar type declarations.
Name two modern features introduced in PHP 8.	JIT (Just-In-Time) compilation and named arguments (also acceptable: match expressions, enums, readonly properties, union types).
Give one real-world example of a major platform using PHP.	WordPress (~40%+ of all websites), Facebook (originally), or Wikipedia's MediaWiki.

9. FAQs

Question	Answer
Is PHP dead / dying?	No. It still powers a huge share of the live web (largely via WordPress), and PHP 8's performance work plus frameworks like Laravel keep it in active commercial use and hiring demand.
Should I learn PHP or Python first?	For this course it doesn't matter — the concepts (variables, loops, functions, databases, security) transfer between languages. PHP has a

Question	Answer
	shorter path from "hello world" to "a working website talking to a database," which is why it's a common first backend language.
Do I need to memorize PHP version numbers for exams?	You should recognize the milestones (PHP 3 = first real language, PHP 5 = OOP, PHP 7 = performance, PHP 8 = modern features) more than exact dates — that's the version table's purpose above.
Why do we say the browser "never sees" PHP code?	Because the web server executes the PHP and only sends the generated output (plain HTML/CSS/JS) to the browser. You can verify this yourself later using "View Page Source" in a browser — you'll never see a line of PHP there, even on a PHP-powered site.

— End of Class 1 Notes —